

The Downward Step: What Gradient Descent Assumes About Learning

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A hill, a ball, and an arrow pointing downward: introductory accounts of gradient descent often picture learning as a journey toward the bottom of a landscape. The image is persuasive because it makes improvement look like a feature of the terrain itself. Yet the landscape is not discovered in the way a mountain is discovered. Someone chooses what to measure, which mistakes matter, and how those mistakes become heights. Gradient descent is a powerful method for adjusting a model's parameters to reduce a mathematical objective; it is not a method for discovering what an objective ought to be. Read as a technical artifact rather than merely a procedure, its famous update rule makes an implicit argument: once a problem has been expressed as a differentiable loss, purposeful change can be organized as a sequence of locally justified steps. The rule's precision is real. So is the authority it quietly grants to decisions made before optimization begins.

In its simplest form, the construction can be written as follows:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

The symbol θ names a collection of adjustable parameters, such as the weights in a predictive model; t marks the current iteration. L is the loss assigned to a particular setting of those parameters, ∇L is its gradient, and η is a positive learning rate. At each step, the algorithm subtracts a scaled gradient from the current parameters. For a differentiable loss, the negative gradient points in the direction of steepest local decrease under the usual Euclidean measure of distance. With a sufficiently small step, this direction will ordinarily reduce the loss nearby; a careless step can instead overshoot. Nothing in the notation specifies what is being predicted, who is affected, or why a lower number should count as progress (Goodfellow, Bengio, and Courville 2016). Those absences are not errors in the mathematics. They are what allow the same rule to travel between applications.

Consider first the modest-looking θ . It identifies what the system is permitted to change. During an ordinary gradient-descent update, the model can revise weights, but it does not revise its own training labels, question the legitimacy of a dataset, or renegotiate the task assigned by its designers. The apparently open-ended word “learning” therefore names a bounded form of adaptation. An institution using a model to sort applicants might let its parameters move millions of times while leaving untouched the category “qualified applicant.” That category may embody an older institutional judgment. In the equation, however, such judgments appear only indirectly, as features and labels already built into the world the parameters must fit. The algorithm has plasticity within the frame and no authority over the frame itself.

The loss function L gives this frame its direction. Calling its output a “loss” sounds descriptive, as though the procedure simply detects a preexisting quantity of wrongness. But loss is constructed. A squared-error function makes large numerical deviations count disproportionately; a classification loss may punish confident incorrect predictions especially strongly. These choices have excellent mathematical reasons and different practical consequences (Google for Developers, n.d.). They also establish which differences are commensurable. To obtain one number, many distinct events must be made comparable: an error on one person’s record can offset an improvement on another’s. The scalar is convenient because it makes a gradient calculable. Its convenience should not be confused with an account of whose errors are bearable or whose needs are urgent.

When a loss is averaged over training examples, another decision becomes hard to see. The arithmetic mean treats the sample as the field over which success will be measured. A poorly represented group may contribute little to the aggregate gradient, even when errors concerning that group are severe. This outcome is not inevitable: designers can reweight examples, change the objective, or impose additional constraints. But each repair requires an explicit choice about what deserves attention. Suresh and Gutttag (2021) distinguish harms arising at multiple stages of the machine-learning life cycle, including representation, measurement, evaluation, and deployment. Their framework helps locate what the update rule cannot diagnose. If the sample excludes certain people or the labels misdescribe their situation, improving the fit to that sample is not the same thing as repairing the exclusion.

The subtraction sign deserves equally close attention. It makes the rule look self-correcting: find the slope, reverse it, and descend. In a smooth landscape this is locally rational, but its rationality is conditional. “Down” means down relative to L , not necessarily toward truth, justice, or a useful decision. A valley may be a local minimum rather than the best available solution, and a flat region may frustrate progress. More fundamentally, loss on a training set is only a stand-in for performance in cases the model has not seen. As Goodfellow, Bengio, and Courville (2016) emphasize, machine learning often optimizes a tractable surrogate in the hope of improving a different quantity of ultimate interest. The optimization problem and the human problem are related by an assumption, not by an identity.

The learning rate η introduces a further, easily overlooked agent: the person or process choosing how aggressively change should occur. A small η can make training painstakingly slow; a large one can produce instability rather than descent (Google for Developers, n.d.). Thus even the apparently natural metaphor of following a hill requires governance. The gradient supplies a direction only at the present location. It cannot, by itself, specify an appropriate stride, a stopping rule, or how to respond when the ground changes. These are engineering choices, not merely decorative settings. The rule also privileges local evidence: it recommends what to do next from the slope here, not from an independent understanding of the

entire landscape. Its procedural humility is a strength in enormous parameter spaces, but it is not omniscience.

That limitation matters because a successful descent can conceal incompatible solutions. D’Amour and colleagues (2022) describe “underspecification”: substantially different predictors may perform similarly on standard tests yet behave differently in deployment. A low reported loss cannot tell us which cues a model relied upon if multiple strategies earn the same score. Nor is the path uniquely fixed in contemporary practice. Mini-batch training estimates gradients from subsets of data, while initialization, sampling, and other training choices can affect the result. Gradient descent should therefore not be imagined as a moral compass or an infallible route to a single correct answer. It is a method for making a particular sort of numerical progress within a constructed problem.

A case from healthcare reveals the stakes of that construction. Obermeyer and colleagues (2019) studied a widely used system that helped identify patients for additional care. Its target was future healthcare spending, treated as a proxy for medical need. Yet because less money was spent on Black patients with comparable health needs, cost was not an impartial measure of illness. The researchers found that Black patients at the same risk score were, on average, sicker than White patients. The failure lay in the choice of target: accurately forecasting spending could still allocate help unfairly. The study does not establish that a particular gradient-descent implementation caused the disparity; it shows why even a flawlessly executed optimization procedure cannot rescue an inadequately specified goal. A number may be predicted correctly while the human question has been answered badly.

This example clarifies who appears in the update rule as a user and who appears as an object. The imagined user is someone authorized to define loss, collect data, set η , and decide when training is good enough: a developer, organization, or technical operator. The imagined object is the adjustable model, and, at one remove, the people converted into records through which the model learns. Those people may later experience the decisions, but their ability to contest the categories is not represented by any symbol. Selbst and colleagues (2019) warn that technical abstraction can sever an apparently self-contained fairness problem from the institutions and relationships that give it meaning. Read beside their argument, gradient descent’s portability becomes double-edged: it is useful precisely because it abstracts from context, but the same abstraction makes context easier to mistake for irrelevant background.

There is a second exclusion, built into the requirement for a usable gradient. An experience has no direct influence on the update unless the system translates it into the chosen computational objective or some other mechanism affecting training. Humiliation, dignity, distrust, and the possibility of appeal do not arrive as derivatives. They can be partially represented by proxies, annotations, additional objectives, or constraints; none is forbidden by mathematics. Yet every

such translation asks whose description is accepted and what remains unmeasured. The formula is consequently not an argument that only quantifiable things matter. It is an argument that, for this form of learning to act on something, someone must first make it operational in a particular way. The excluded material does not cease to exist when it disappears from the equation.

A defender of gradient descent might reasonably object that these criticisms target bad goals rather than the optimization method. That distinction must be maintained. The same update rule can train a model with a better target, a differently weighted loss, or constraints intended to reduce harmful disparities. It has enabled genuine achievements that would be impossible to obtain by manually setting every parameter. Nor does every application carry the moral stakes of healthcare or hiring. Nevertheless, the distinction between method and goal does not make the method culturally invisible. A technique that works after ends have been reduced to a smooth numerical surface encourages practitioners to look for problems that can take that form. It offers an attractive picture of responsibility as improved metrics, even though selecting, measuring, and contesting those metrics remain separate responsibilities.

The small minus sign, then, is both the rule's triumph and its limit. It coordinates astonishingly many revisions without needing an explicit account of the world at every step. But its certainty concerns direction on a specified surface, not the worth of that surface. To read gradient descent closely is not to accuse an equation of harboring secret intentions. It is to notice a division of labor that its elegance can disguise: humans and institutions decide what counts as error, while the algorithm becomes exceptionally good at reducing it. The more persuasive the descent looks, the more important it is to ask who drew the landscape, who can change it, and who must live at the bottom.

References

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